An Artificial Bee Colony Algorithm Based Approach to the Constrained p-Center Problem

Anil Panchumarthi and Alok Singh
Department of Computer & Information Sciences
University of Hyderabad, Hyderabad 500 046, India
Email: anil.panchumarthi89@gmail.com, alokcs@uohyd.ernet.in

***Abstract*—p-center problem has been studied widely in operations research and computational geometry. It is a facility location problem. Here the goal is to find the best positions of p centers in a plane consisting of n demand points such that the maximum distance of any point from its nearest center is minimized. Constrained p-center problem is an extension of p-center problem where there are certain restrictions on placement of centers. This problem is more suitable to model real world applications than its unconstrained counterpart because of the practical restrictions on the placement of facilities, for example ATMs or gas stations can not be placed in the middle of a lake or a pond. This paper describes an artificial bee colony (ABC) approach for constrained p-center problem. Computational results show the effectiveness of our approach.**

***Index Terms*—Artificial Bee Colony Algorithm; Constrained p-Center Problem; Constrained Optimization; Facility Location; Swarm Intelligence**

I. Introduction

p-center problem is a well known facility location problem. Suppose n points are given in a plane, then p-center problem consists in finding the location of p-centers in the plane such that the maximum distance of any given point in the plane from its nearest center should be minimum. Figure 1 shows sample representation of p-center problem with $p = 1$. The problem is $\mathcal{NP}$-Hard when parameter p is a part of input [1]. p-center problem can be defined in both discrete and continuous domains. However, there is no universally acceptable definition of continuous domain with respect to this problem. In computational geometry area, location problem is considered continuous if the set of all possible locations of centers is infinite ([4], [5], [6]). In the field of operations research, when points set is infinite then the problem is called continuous problem [8]. In this paper like in [14], we assume that points set is discrete, but the set of possible locations for centers is infinite. Formally, p-center problem can be defined as follows: given a set S=$\{s_1,s_2,\ldots,s_n\}$ of n points in a plane, p-center problem seeks the location of p centers c_1, $c_2,\ldots$, c_p such that the maximum distance of any given point from its closest center is minimized, that is

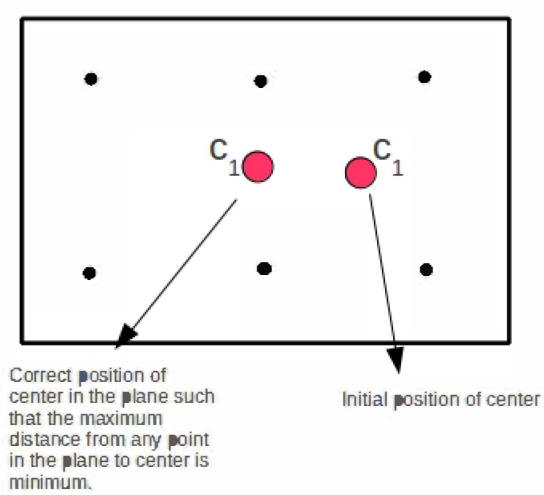

Fig. 1. Sample representation of points on a plane with centers

$$Min_{1\leq j\leq p}\{Max_{1\leq i\leq n}\{Min_{1\leq j\leq p}\ d(p_i,c_j)\}\}$$

where $d(a,b)$ is the euclidean distance between two points a and b which are located at (x_a,y_a) and (x_a,y_a) respectively, i.e.,

$$d(a,b)=\sqrt{(x_a-x_b)^2+(y_a-y_b)^2}$$

The best algorithm known for above problem takes $\mathcal{O}(n^{\sqrt{p}})$ time [15], but for small p values there are better algorithms. When p is one the corresponding problem is called minimum covering circle problem which has been discussed in [16], [17]. By using a parametric searching algorithm Megiddo [18] has solved minimum covering circle in linear time.

Several versions of p-center problem were discussed in [7]. To solve different versions of p-center problem, many heuristic and approximation algorithms have been proposed ([5], [9], [10], [11]). The p-center problem on graphs and trees was studied by Caruso et al. [2] and Cheng et al. [3]. These studies are performed with the purpose of solving a variety of real life location problems.

Constrained p-center problem is an extension of p-center problem where centers can not be placed anywhere in the plane, i.e., there are certain restrictions on placement of centers. This model is more closed to real world as facilities like ATM or fuel station can not be placed in a lake or a pond or on a hill top. In addition to facility location, constrained version has been used in data mining, clustering, image processing etc. ([19], [20]). Davoodi et al. [14] proposed an approach to solve this problem by using a non-deterministic heuristic algorithm

based on Voronoi diagrams(VD).

In this paper, we have proposed an artificial bee colony (ABC) algorithm for the constrained p-center problem. We have also developed a modified version of K-Means algorithm to solve this problem. Computational results show the effectiveness of our ABC approach.

The remainder of this paper is organized as follows: Section II provides an overview of the artificial bee colony algorithm. Section III describes our ABC approach for constrained p-center problem. Section IV presents modified K-Means algorithm. Computational results are presented in section V. Finally, section VI contains some concluding remarks and directions for future research.

II. Overview of Artificial Bee Colony Algorithm

Artificial Bee Colony (ABC) algorithm is a swarm intelligence technique which is inspired by the foraging behavior of natural honey bee swarm. Since the development of first ABC algorithm by Karaboga [21] in 2005, several variations of basic algorithm have been reported in the literature ([22], [23], [24], [25], [26], [27], [28], [29], [30], [31]).

Foraging bees are divided into three groups– employed bees, onlooker bees and scout bees. Employed bees are responsible for collecting the nectar from food sources and sharing information about their food sources with the onlooker bees. Onlooker bees wait in the hive for employed bees to share information about food sources' location and nectar content. Scout bees search for new food sources in the vicinity of the hive. Employed bees share the information with onlooker bees about their respective food sources by dancing in a designated area of the hive called dance area. The nature and duration of the dance depends on nectar content, direction and distance of the food source. Onlooker bees view several dances before selecting a food source. Onlooker bees select food sources probabilistically where the probability of selection of a food source is proportional to its nectar content. As a result, good food sources get more onlookers than the worse ones. Whenever an onlooker or a scout bee finds a food source, it becomes employed. Whenever a food source is exhausted, its associated employed bees abandon it and become either scouts or onlookers. Therefore, employed and onlooker bees can be perceived as performing the job of exploitation, whereas scout bees can be perceived as performing the job of exploration. Moreover, good food sources are exploited more heavily in comparison to worse ones as they attract more onlookers. Motivated by this foraging behavior Karaboga [21] proposed the ABC Algorithm. In essence, ABC algorithm is based on the fact that in the neighborhood of good solutions, the chances of finding even better solutions are higher and therefore, neighborhood of good solutions should be explored more thoroughly than the worse ones.

In ABC algorithm, each food source represents a candidate solution to the problem under investigation and the nectar content of a food source corresponds to the fitness of the solution being represented by that food source. Karaboga [21] also divided the colony of artificial bees into same three groups, viz., employed, onlookers and scouts with similar functions. However, the number of employed bees is always taken to be equal to the number of food sources. Usually (but not always), the number of onlooker bees are also taken to be equal to the number of employed bees. The employed bee associated with an exhausted food source always becomes a scout (but never an onlooker). In the ABC algorithm, the action of a scout bee is modeled by generating a new food source (solution) randomly and associating the scout bee in consideration with this new food source to make it employed again. We will use food source & solution and nectar content & fitness interchangeably from now onwards.

The ABC algorithm starts by associating each employed bee with a randomly generated food source. After this initialization phase, ABC algorithm follows an iterative process. In each iteration, each employed bee determines a new food source in the neighborhood of its currently associated food source and evaluates the nectar content of this new food source. If the nectar content of this new food source is higher than that of its currently associated food source, then this employed bee moves to the new food source abandoning the old one, otherwise it remains associated with the old one. The manner in which a food source in the neighborhood of a particular food source is determined is problem dependent. When all employed bees are done with this process, they share information about their respective food sources with onlookers. Onlookers tend to select the food sources probabilistically according to their nectar content (fitness). The exact method of selection varies from one implementation of ABC algorithm to another. The roulette wheel selection method and binary tournament select method are the two most commonly used selection methods. Once all onlookers select their food sources, each of them determines a new food source in the neighborhood of its selected food source in the same manner as employed bees and evaluates the nectar content of this new food source. The best food source among all the neighboring food sources determined by the onlookers associated with a particular food source i and the food source i itself, becomes the new location of the food source i in the next iteration. If the solution represented by a food source i does not improve over a specified number of iterations then food source i is assumed to be completely exhausted and its associated employed bee abandons it to become scout. This scout is immediately made employed by associating it with a new randomly generated food source. Once the new locations of all food sources are determined, next iteration of ABC algorithm starts. The whole process is repeated again and again as long as termination condition remains unsatisfied. Throughout the ABC algorithm, the best solution found so far is tracked and this solution is returned as the solution found by the ABC algorithm at the end.

For a comprehensive survey on ABC algorithm and its applications, interested readers may refer to [32].

III. ABC APPROACH FOR CONSTRAINED p-CENTER PROBLEM

In this section, we present our ABC approach for the constrained p-center problem. Following subsections describe the main features of our ABC approach.

A. Initial solution generation

Each initial solution is generated by placing each of the p centers randomly within the permissible range of coordinates in the plane. If a center got placed in a restricted area then we replace this location with its nearest location outside the restricted area.

B. Process of selecting a food source for an onlooker bee

We have used the probabilistic binary tournament selection method to select a food source for each onlooker bee. In this method, two food sources are chosen uniformly at random from all food sources and the better of the two is selected with probability π_{better}, otherwise the worse of the two is selected.

C. Generation of neighboring solution

The method used here is derived from the method used in [24] to generate neighboring solutions. In order to generate a neighboring solution e'_i for a solution e_i, we select another solution e_k (different from e_i) randomly. Then we select a center, say j, randomly. We calculate the difference (in the sense of any one of the coordinates which is again chosen randomly) between the jth center in e_i and e_k. We multiply this difference with a random constant $\phi_{i,j} \in [-1, 1]$. This result is added to the location of jth center in e_i. If the resulting value is not in the permissible range, then this value is replaced with a random value in the range. More formally, the neighboring solution generation is based on following formula [24]:

$$e'_{ij} = e_{ij} + \phi_{ij}(e_{ij} - e_{kj}),$$

$i \in \{1, 2, \ldots, n_e\}$, $k \in \{1, 2, \ldots, n_e\}$ and $j \in \{1, 2, \ldots, p\}$
Here n_e is the number of employed bees and e_{ij} refers to the jth center of solution e_i. Please note that j is chosen randomly. If the resulting location falls in a restricted area then replace that location with its nearest location outside the restricted area. Figure 2 illustrates this process pictorially.

Unlike the traditional artificial bee colony algorithms where we generate only one neighboring solution for an employed or onlooker bee, here we are generating p such solutions and using the best solution among these p solutions for further comparison and possible replacement.

D. Other Features

If a solution does not improve for a specified number of iterations then that solution is replaced with a randomly generated solution. This solution is generated in the same manner as initial population.

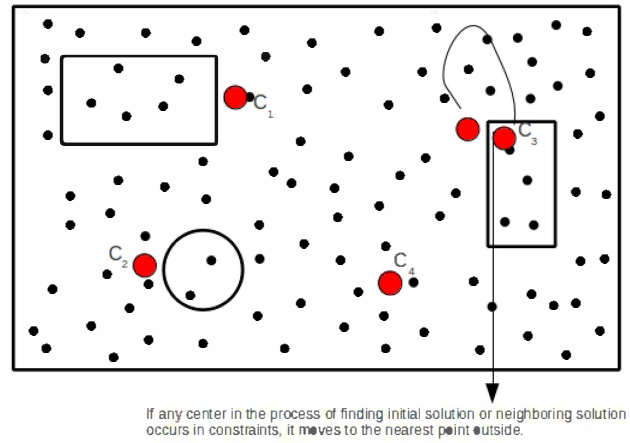

Fig. 2. Movement of the newly generated center when it lies in a restricted area

IV. K-MEANS ALGORITHM

To compare results of our ABC approach, we have developed a modified K-Means algorithm for constrained p-center problem which is described below:

1) Generate p centers randomly. Same process as used in initial solution generation of ABC algorithm is used here.
2) Find the nearest center for every point in the plane and group the points according to their nearest center, i.e., points having the same nearest center belong to the same group. This will result into p groups.
3) Compute mean of each group and replace centers with these means. If a center falls in a restricted area then replace this center with nearest location outside the restricted area.
4) Repeat steps 2–3 until step 3 fails to produce any change in centers.
5) Calculate the fitness of the final solution and report the result.

V. RESULTS AND DISCUSSIONS

We have used C programming language for implementing ABC algorithm and modified K-Means algorithm for constrained p-center problem. These approaches have been executed on an Intel core i3 system with 4GB RAM running at 2.83 GHz. For our ABC approach, we have used a population of 50 employed bees and 100 onlooker bees. If an employed bee solution does not improve over 100 iterations then we are replacing that solution with a randomly generated solution. We have used $\pi_{better} = 0.9$. We have allowed our ABC approach to execute for 2000 iterations. All these parameter values are set to their respective values after a large number of trials involving several instances. These parameter values are just one set of parameter values among many possible sets which give good results on most of the instances, and are in no way optimal parameter values for all instances.

To test our ABC approach, we have considered a rectangular plane with lower left coordinate (20, 50) and upper right

TABLE I
ABC RESULTS

p	number of points - n								
	20	30	40	50	80	100	200	500	1000
2	162.89	178.77	178.12	177.75	195.27	202.31	211.67	217.23	222.42
5	96.17	105.38	105.38	108.37	120.73	126.08	127.86	136.13	139.28
8	70.89	84.19	88.37	91.82	94.23	101.69	102.31	106.34	113.70
10	61.43	67.71	70.61	72.29	75.14	87.26	98.64	96.17	101.82
15	31.98	39.12	43.67	52.43	66.17	64.78	75.27	82.62	86.23
20	0	39.52	39.52	51.85	59.21	61.34	66.08	69.39	79.78
30	0	0	36.21	46.88	49.24	50.74	57.22	66.27	72.86
40	0	0	0	39.43	46.14	49.26	53.82	57.86	61.43
50	0	0	0	0	41.73	46.31	48.61	49.73	54.34

coordinate (599, 399). We have considered two rectangles and one circle in the plane as restricted area, i.e., centers can not be placed inside these. The first rectangle has (90, 320) and (170, 360) respectively as lower left and upper right coordinates. The second rectangle has (500, 150) and (530, 210) respectively as lower left and upper right coordinates. The circle is centered at (150, 150) with a radius of 30. We have taken the number of points n in { 20, 30, 40, 50, 80, 100, 200, 500, 1000} and number of centers p in { 2, 5, 8, 10, 15, 20, 30, 40, 50}. In our approaches, we have used a pre-processing step to identify those cases where $p \geq n$. For these cases we have an optimal solution with cost 0 as n centers out of p centers can be placed exactly at n points and remaining centers can be placed randomly. For such cases pre-processing phase only outputs the result and ABC/K-Means algorithm is not applied.

Tables I and II present our results. Table I presents the result of our ABC approach whereas table II does the same for K-Means algorithm. These tables clearly show the superiority of our ABC approach as it is able to find better solutions for most of the instances. For some instances ABC algorithm even improved the results obtained through modified K-Means algorithm by more than 20%. For largest instances with 1000 points ABC algorithm always obtained better results than K-Means algorithm.

VI. Conclusion and future work

In this paper, we have presented an artificial bee colony algorithm based approach for constrained p-center problem. We have also developed a modified version of K-Means algorithm for constrained p-center problem. This problem find practical applications in facility location in those situations where facilities can not be located arbitrarily. Computational results show the effectiveness of our ABC approach.

As a future work, we would like to incorporate a local search within our ABC approach to further improve its quality. We would also like to explore different neighboring solution generation techniques to investigate their effect on the performance of ABC algorithm. In a similar way other swarm intelligence techniques can also be developed for this problem.

Acknowledgment

The second author would like to thank the Department of Science & Technology, Government of India for its financial support in the form of a research grant to carry out this research work.

References

[1] R.J. Fowler, M.D. Paterson, D.L. Tanimoto, "Optimal packing and covering in the plane are NP-complete," *Information Processing Letters*, vol. 12, pp. 133-137, 1987.

[2] C. Caruso, A. Colorni, L. Aloi, Dominant, "an algorithm for the p-center problem," *European Journal of Operational Research*, vol. 149, pp. 53-64, 2003.

[3] T.C.E. Cheng, L. Kang, C.T. Ng, "An improved algorithm for the p-center problem on interval graphs with unit lengths," *Computers and Operations Research*, vol. 34, pp. 2215-2222, 2007.

[4] Z. Drezner, "The p-center problem – heuristics and optimal algorithms," *Journal of the Operational Research Society*, vol. 35, pp. 741-748, 1984.

[5] A. Suzuki, Z. Drezner, "The p-center location problem in an area", *Location Science*, vol. 4, pp. 69-82, 1996.

[6] A. Suzuki, Z. Drezner, "The minimum equitable radius location problem with continuous demand," *European Journal of Operational Research*, vol. 195, pp. 17-30, 2009.

[7] P.K. Agarwal, M. Sharir, "Efficient algorithms for geometric optimization," *ACM Computing Surveys*, vol. 30, pp. 412-458, 1998.

[8] M.S. Daskin, *Network and Discrete Location: Models, Algorithms and Applications*, John Wiley, New York, 1995.

[9] S. Elloumi, M. Labbe, Y. Pochet, "A new formulation and resolution method for the p-center problem," *INFORMS Journal on Computing*, vol. 16, pp. 84-94, 2004.

[10] D. Hochbaum, D.B. Shmoys, "A best possible heuristic for the k-center problem," *Mathematics and Operations Research*, vol. 10, pp. 180-184, 1985.

[11] D.S. Hochbaum, A. Pathria, "Generalized p-center problems: complexity results and approximation algorithms," *European Journal of Operational Research*, vol. 100, pp. 594-607, 1997.

[12] H. Wei, A.T. Murray, N. Xiao, "Solving the continuous space p-centre problem: planning application issues," *IMA Journal of Management Mathematics*, vol. 17, pp. 413-425, 2006.

[13] M.E. Dyer, A.M. Frieze, "A simple heuristic for the p-center problem," *Operations Research Letters*, vol. 3, pp. 285-288, 1985.

[14] Mansoor Davoodi, Ali Mohades, Jafar Rezaei, "Solving the constrained p-center problem using heuristic algorithms", *Applied Soft Computing*, vo. 11, pp. 3321-3328, 2011.

[15] R.Z. Hwang, R.C.T. Lee, R.C. Chang, "The slab dividing approach to solve the Euclidean P-center problem", *Algorithmica*, vol. 9, pp. 1-22, 1993.

[16] J. Elizinga, D.W. Hearn, "Geometrical solution to some minimax location problems", *Transportation Sciences*, vol. 6, pp. 379-394, 1972.

[17] M.I. Shomos, D. Hoey, "Closest-point problems", *Proceedings of 16th Annual IEEE Symposium on Foundations of Computer Science*, pp. 151-162, 1975.

[18] N. Megiddo, "Linear-time algorithms for linear programming in R3 and related problems," *SIAM Journal on Computing*, vo. 12, 759-776, 1983.

[19] M.L. Brandeau, D.S. Chiu, "An overview of representative problems in location research," *Management Sciences*, vo. 35, pp. 645-674, 1989.

[20] J.M. Cadenas, M.J. Canos, M.C. Garrido, C. Ivorra, V. Liern, "Soft-computing based heuristics for location on networks: The p-median problem," *Applied Soft Computing*, vol. 11, pp. 1540-1547, 2011.

TABLE II
K-MEANS RESULTS

p	number of points - n								
	20	30	40	50	80	100	200	500	1000
2	182.16	186.87	192.38	196.13	211.48	213.21	222.42	235.28	239.23
5	109.37	121.17	126.31	137.21	143.79	148.52	157.18	166.12	168.77
8	92.08	95.37	98.11	111.64	121.53	124.87	128.62	130.37	134.17
10	76.67	89.21	94.36	104.63	110.13	114.53	118.93	122.27	125.32
15	42.29	47.61	59.21	70.61	79.37	82.32	96.12	88.76	92.16
20	0	40.38	43.69	62.47	66.82	71.18	73.91	77.68	82.35
30	0	0	33.87	46.27	57.23	62.31	67.46	69.19	79.67
40	0	0	0	32.64	43.24	51.81	53.72	56.29	64.37
50	0	0	0	0	36.47	41.53	43.07	47.16	58.71

[21] D. Karaboga, *An idea based on honey bee swarm for numerical optimization,* Technical Report TR06, Computer Engineering Department, Erciyes University, Turkey, 2005.

[22] B. Basturk, D. Karaboga, "An artificial bee colony (ABC) algorithm for numeric function optimization," *Proceedings of the IEEE Swarm Intelligence Symposium*, Indianapolis, IN, USA, May 12-14, 2006.

[23] D. Karaboga, B. Basturk, "A powerful and efficient algorithm for numerical function optimization: artificial bee colony (ABC) algorithm," *Journal of Global Optimization*, vol. 39, pp. 459-471, 2007.

[24] D. Karaboga, B. Basturk, "On the performance of artificial bee colony (ABC) algorithm," *Applied Soft Computing*, vol. 8, pp. 687-697, 2008.

[25] D. Karaboga, B. Akay, "A modified Artificial Bee Colony (ABC) algorithm for constrained optimization problems," *Applied Soft Computing*, vol 11, pp. 3021-3031, 2011.

[26] A. Singh, "An Artificial Bee Colony algorithm for the leaf-constrained minimum spanning tree problem," *Applied Soft Computing*, vol. 9, pp. 625-631, 2009.

[27] S. Sundar, A. Singh, "A swarm intelligence approach to the quadratic multiple knapsack problem," *Proceedings of the* 17th *International Conference on Neural Information Processing*, LNCS 6443, pp. 626-633, Springer-Verlag, Berlin, 2010.

[28] S. Sundar, A. Singh, "A swarm intelligence approach to the quadratic minimum spanning tree problem," *Information Sciences*, vol. 180, pp. 3182-3191, 2010.

[29] S. Sundar, A. Singh, "A Swarm intelligence approach to the early/tardy scheduling problem," *Swarm and Evolutionary Computation*, vol. 4, pp. 25-32, 2012.

[30] Q.-K. Pan, M.F. Tasgetiren, P.N. Suganthan, T.J. Chua, "A discrete artificial bee colony algorithm for the lot-streaming flow shop scheduling problem," *Information Sciences*, vol. 181, pp. 2455-2468, 2011.

[31] W.-F. Gao, S.-Y Liu, "A modified artificial bee colony algorithm," *Computers & Operations Research*, vol. 39, pp. 687-697, 2012.

[32] D. Karaboga, B. Gorkemli, C. Ozturk, N. Karaboga, "A comprehensive survey: artificial bee colony (ABC) algorithm and applications," *Artificial Intelligence Review*, In Press, doi:10.1007/s10462-012-9328-0, 2012.